Does a model that predicts well also fly well?

World models under test

JEPA: `full`, `no_cross`, `flat`

W-ar: autoregressive state model

PETS: probabilistic ensemble

3 seeds each at 500 and 2,000 episodes

references

physics; physics + mass estimation

oracle U: the true simulator

Shared planner, fixed (MPPI, 1.5 s ahead)

512 candidate plans

state, 0–200 ms old

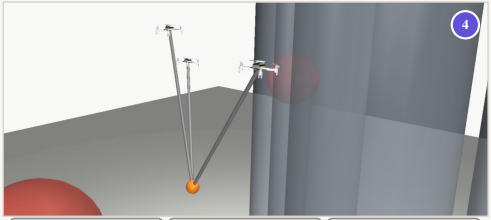
world model under test 2

predicted futures 1

MPPI-weighted plan 3

setpoints, 10 Hz

Task: three Crazyflie 2 carry a slung load



hidden load mass | 0–200 ms delay | moving obstacles

Four evaluation layers: overview here, definitions in Fig. 4

1 Prediction
Same plan:
load trajectory error

2 Counterfactual
Change one UAV:
cross-agent effect error

3 Decision
Regret of the weighted execution plan

4 Mission
Success rate J over 400 test episodes

Pre-registered test
Does L_2 add information about J beyond L_1?
30 learned models.